ZIMSEC
For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS **6032/3**
PAPER 3 THEORY 2 hours 30 minutes

JUNE 2026 SESSION

Additional materials:
Answer booklet
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.
Write your answers on the separate answer booklet provided.
If you use more than one booklet, fasten the booklets/sheets together.
All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 13 printed pages.

DATA

speed of light in free space	$c = 3.00 \times 10^8 ms^{-1}$
permeability of free space	$\mu_o = 4\pi \times 10^{-7}$ Hm^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ Fm^{-1} $(1/4\pi\varepsilon_0 = 8.99 \times 10^9$ $mF^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
the Planck constant	$h = 6.63 \times 10^{-34}$ Js
unified atomic mass unit	$1\ u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ $JK^{-1}mol^{-1}$
the Avogadro constant	$N_A = 6.02 \times 10^{23} mol^{-1}$
the Boltzmann constant	$k = 1.38 \times 10^{-23}$ JK^{-1}
gravitational constant	$G = 6.67 \times 10^{-11} Nm^2kg^{-2}$
acceleration of free fall	$g = 9.81$ ms^{-2}

FORMULAE

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p\,\Delta V$
gravitational potential	$\emptyset = -Gm/r$
hydrostatic pressure	$p = \rho gh$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_o \cos \omega t$
	$v = \pm\omega\sqrt{(x_o^2 - x^2)}$
Doppler effect	$f_o = \frac{f_s v}{v \pm v_s}$
Attenuation of x-rays	$I = I_o e^{-\mu x}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy of charged capacitor	$W = \frac{1}{2}QV$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_o \sin\omega t$
radioactive decay	$x = x_o \exp(-\lambda t)$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$

$a = -w^2 x_o \sin wt$

[Turn over

Answer question 1 and any other 3 from the remaining questions.

√ **1.** **(a)** **(i)** Define a *base unit* and give one example.

(ii) The volume, V, of liquid that flows through a pipe in time, t, is given by the expression: $\frac{V}{t} = \frac{\pi P r^4}{8Cl}$, where P is the pressure of the liquid, r is the radius of the pipe with length, l, and C is a constant with units $kgm^{-1}s^{-1}$.

Use base units to check the homogeneity of the equation. [5]

(b) **(i)** State the relationship between *systematic error* and *accuracy*.

(ii) In a laboratory, two students were given a wire to measure its diameter. **Table 1.1** shows two sets of measurements that were obtained.

Table 1.1

Student A	**Student B**
d/mm	d/mm
0.38	0.40
0.36	0.41
0.40	0.39
0.45	0.42
0.42	0.38

1. The diameter of the wire was 0.40 mm. Compare the measurements of student A and student B in terms of accuracy.

2. Compare the measurements of student A and student B in terms of precision.

3. Discuss how random errors might have affected student A's results.

[7]

(c) **(i)** Define *torque of a couple.*

(ii) A wooden bar of length 60 cm is pivoted at its centre. Equal and opposite forces of magnitude 3.0 N are applied to the ends of the bar as shown in **Fig 1.1.**

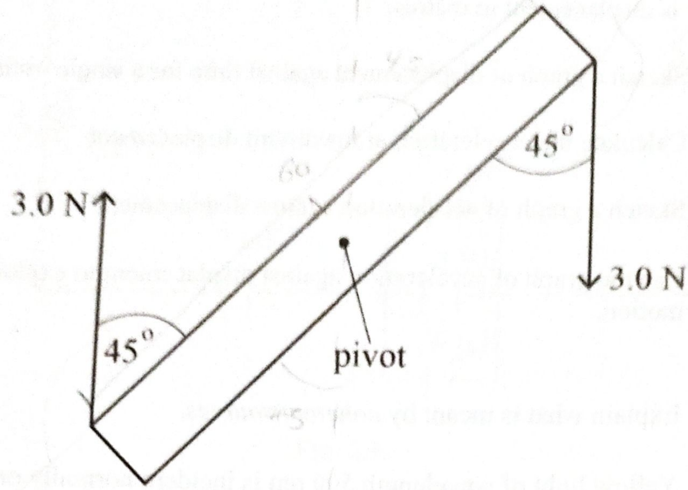

Fig 1.1

$\Sigma t = 0$

(ii) Calculate the torque of a couple on the wooden bar. [3]

(d) A stone of mass 0.5 kg attached to a string of length 0.6 m is whirled in a vertical circle at a constant speed. The maximum tension in the string is 20 N.

(i) Calculate the

1. speed of the stone,

2. minimum tension in the string,

3. time it takes to cover a range of 2 m.

(ii) Explain the origin of the centripetal force in **(d)**.

[7]

(e) **(i)** Define *gravitational field strength.*

(ii) Sketch a graph showing how gravitational field strength (g) varies with distance (r) from the centre of the earth.

[3]

$2 = \frac{1}{2} a t^2$

$2 = \frac{1}{2}(9,81)t^2$

$s = \frac{1}{2}(u+v)t$

$2 = \frac{1}{2}(3)t$

$2 = 1,5t$

$1,34 = t$

2. (a) The vibrations of a component in a machine is represented by the equation:

$$x = 0.3 \times 10^{-4} \sin(240\pi t).$$

where x is displacement in metres.

(i) Sketch a graph of displacement against time for a single oscillation.

(ii) Calculate the acceleration at maximum displacement.

(iii) Sketch a graph of acceleration against displacement.

(iv) Use the graph of acceleration against displacement to explain simple harmonic motion. [8]

(b) (i) Explain what is meant by *coherent sources*.

(ii) Yellow light of wavelength 500 nm is incident normally on a double slit with a slit separation of 0.20 mm and 1.8 m from a screen.

Calculate the fringe separation.

(iii) Describe the effect on fringe separation if red light is used.

(iv) Suggest a reason for using red light on traffic robots rather than yellow light. [7]

(c) (i) Explain *attenuation in X-rays.*

(ii) A beam of X-rays of intensity 10 Wm^{-2} with attenuation coefficient of 400 m^{-1} passes through a tissue of thickness 2.00 mm.

Calculate the intensity of the beam after passing through the tissue.

(iii) Sketch a graph of intensity of X-rays against thickness of tissue to show pattern of absorption. [6]

(d) (i) Explain *total internal reflection in fibre optics.*

(ii) Derive the equation $\frac{n_1 \sin i}{\sin C} = n_2$

$$n = \frac{1}{SinC},$$

where n is the refractive index and C is the critical angle. [4]

3. (a) **Fig. 3.1** shows an arrangement of capacitors in a circuit.

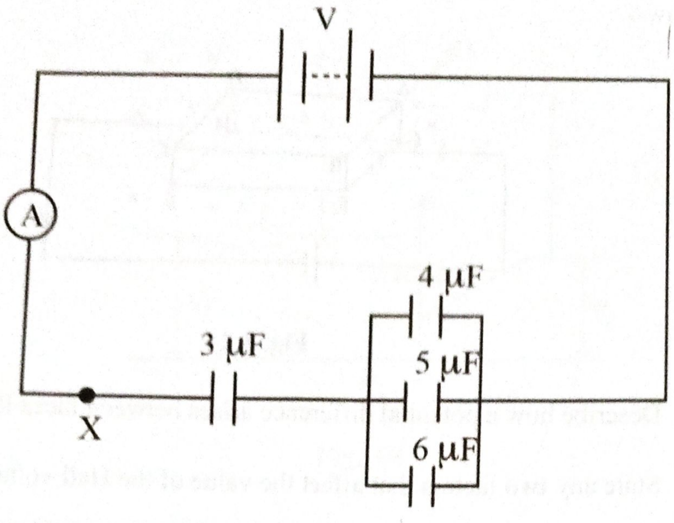

Fig. 3.1

(i) Define *capacitance*.

(ii) Calculate the capacitance of the network.

(iii) Which quantity is the same for the $6\mu F$, $4\mu F$ and $5\mu F$ capacitors.

(iv) State any **two** effects of including a resistor at point X.

[7]

[Turn over

(b) **Fig. 3.2** shows a semiconductor slice in a uniform magnetic field where current I flows.

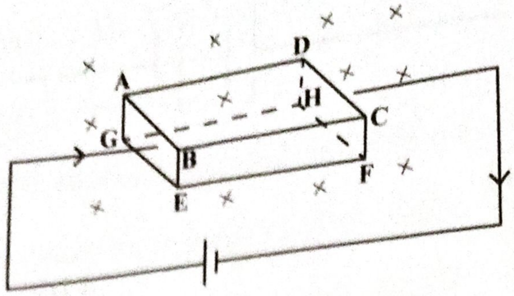

Fig. 3.2

(i) Describe how a potential difference arises between faces BCFE and ADHG.

(ii) State any **two** factors that affect the value of the Hall voltage produced.

(iii) Explain why the electrons will pass through the slice undeflected after some time.

[6]

(c) **Fig. 3.3** shows an electric circuit with resistors and emf sources.

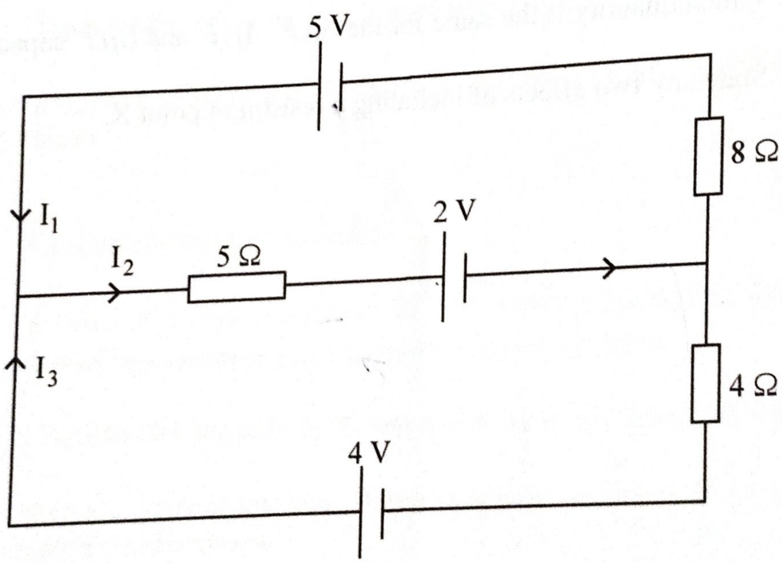

Fig. 3.3

(i) State Kirchhoff's laws and the principles on which they are based on.

(ii) Determine the values of I_1, I_2 and I_3.

[7]

(d) **Fig. 3.4** shows an operational amplifier (opamp) circuit.

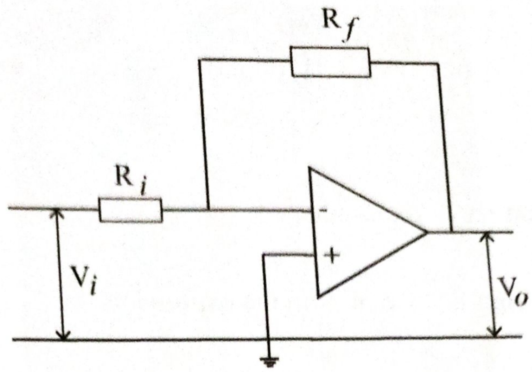

Fig. 3.4

(i) Name the mode of the circuit.

(ii) Describe the virtual earth approximation.

(iii) Derive an expression for the gain of the opamp.

[5]

4. (a) A material subjected to stress may suffer from fatigue and creep.

Give **two** differences between *fatigue* and *creep*. [2]

(b) State what is meant by

(i) *thermodynamic scale,*

(ii) *absolute zero.*

[2]

$V_p = B L \sin\theta$

(c) The pressure exerted by an ideal gas is given by the expression

$$P = \frac{Nm}{3V}\langle c^2 \rangle$$

(i) State

1. what $< c^2 >$ represents,

2. the significance of $\frac{1}{3}$ in the expression.

(ii) Using the expression in **(c)** and the ideal gas equation, show that the average translational kinetic energy of the molecule is proportional to the temperature.

[5]

(d) **Fig. 4.1** shows a venturi meter used to measure the velocity of flow in a pipe. The pressures at A_1 and A_2 are 2.00×10^5 Pa and 1.01×10^5 Pa respectively. The density of water is 1 000 kgm^{-3}.

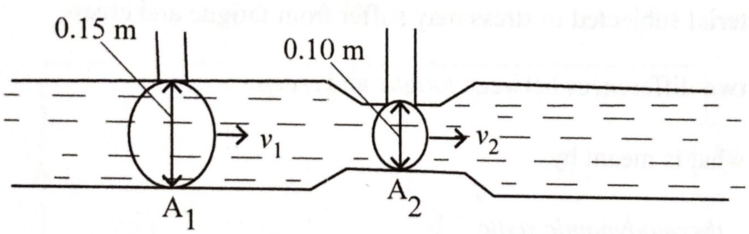

Fig. 4.1

(i) Calculate the velocity at

1. A_1,
2. A_2.

(ii) Using **Fig. 4.1** explain the principle of operation of a venturi meter. [8]

$P_1 + \frac{1}{2}\rho v_1^2 +$

ρ

(e) **Fig. 4.2** shows a pot with boiling water on a hot plate stove.

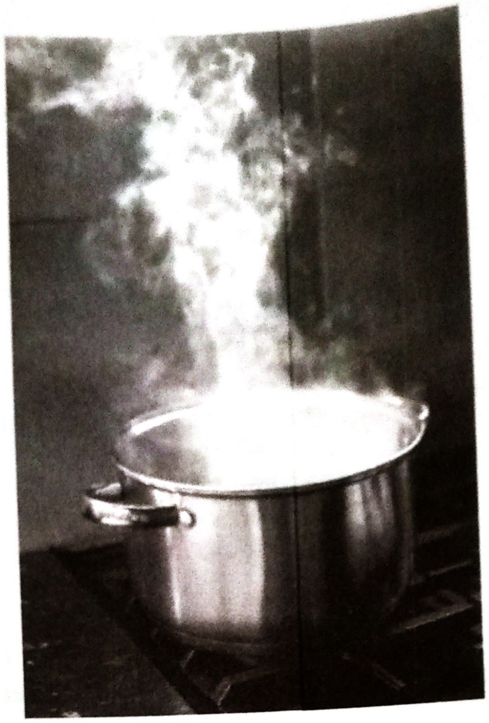

Fig. 4.2

(i) Suggest, with a reason, a suitable material for the handle of the pot.

(ii) Name and explain the process by which heat is transferred from the hot plate stove to the pot and contents.

(iii) Explain the process of heat transfer within the water in the pot.

(iv) A person preparing the food feels the heat.
Explain how heat is transferred to the person. [8]

5. **(a)** **Fig .5.1** shows arrangement of apparatus for the alpha scattering experiment.

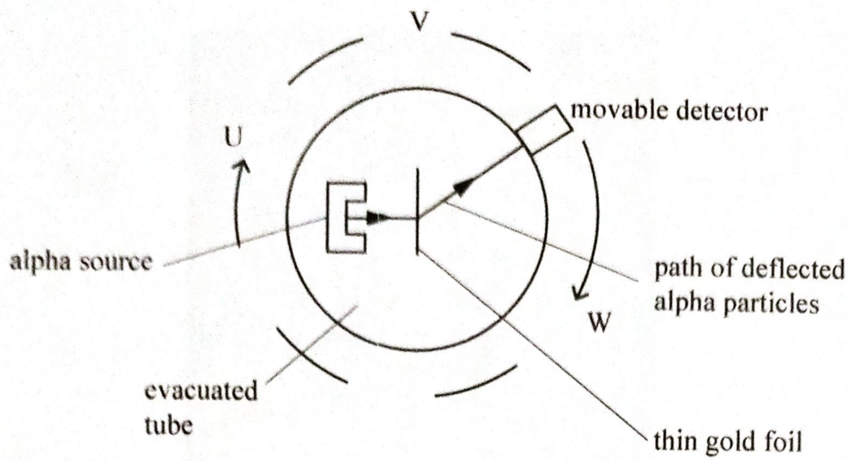

Fig. 5.1

(i) State and explain the observations made when the detector moves along path UVW.

(ii) The alpha particles produce a current of 1.50 mA. Calculate the number of alpha particles passing any point in the beam per second. [8]

(b) Electromagnetic radiation of frequency 1.15×10^{15} Hz is incident on a metal surface with a work function of 2.80×10^{-19} J.

(i) Explain the term *work function.*

(ii) Calculate the

1. energy of a single photon,

2. maximum kinetic energy of electrons,

3. maximum speed of the emitted photo electrons.

[7]

(c) **(i)** Explain the term *bandwidth.*

(ii) State the frequency range of

1. very high frequency (VHF),

2. ultra high frequency (UHF).

(iii) The gain of an amplifier is 50 dB and power input (P_{in}) is 2.0×10^{-7} W. Calculate the power output (P_{out}) of the amplifier.

[6]

(d) **(i)** State what is meant by *background radiation.*

(ii) Describe any **three** environmental importance of background radiation.

[4]

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